Partial Derivatives and the Gradient

ANSWER KEY

First and second partial derivatives

1 For $f(x, y) = x^3y^2 - 4xy + y$, find:

a $f_x \;= 3x^2y^2 - 4y$

b $f_y \;= 2x^3y - 4x + 1$

c $f_{xx} \;= 6xy^2$

d $f_{xy} \;= 6x^2y - 4$

2 For $f(x, y) = e^{xy}\sin(x + y)$, find f_x using the product rule.

$f_x = ye^{xy}\sin(x + y) + e^{xy}\cos(x + y)$.

The gradient and directional derivatives

3 Find the gradient ∇f of $f(x, y) = x^2y + 3y^3$ at the point $(1, 2)$.

$\nabla f = \langle 2xy, x^2 + 9y^2 \rangle$. At $(1, 2)$: $\nabla f(1, 2) = \langle 4, 37 \rangle$.

4 Find the directional derivative of $f(x, y) = x^2 + y^2$ at $(1, 1)$ in the direction of $\mathbf{v} = \langle 3, 4 \rangle$.

$\nabla f = \langle 2x, 2y \rangle$, so $\nabla f(1, 1) = \langle 2, 2 \rangle$. Unit vector: $\hat{\mathbf{v}} = \langle \frac{3}{5}, \frac{4}{5} \rangle$. $D_{\mathbf{v}}f = \langle 2, 2 \rangle \cdot \langle \frac{3}{5}, \frac{4}{5} \rangle = \frac{6}{5} + \frac{8}{5} = \frac{14}{5}$.